Novanex OnAlert™ LED Sign and CAP Feed Integration Overview

Novanex LED Sign and CAP Feed

Integration Overview

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1. Introduction

The Novanex OnAlert™ LED Sign shows your CAP feed notifications in a highly visible manner. The OnAlert sign polls one or two outbound Common Alerting Protocol (CAP) channels at particular URLs and then displays messages posted to those URLs in order of priority.

The *My Displays* tool on the Novanex Support Site provides functionality to send a CAP message to an OnAlert sign using the Message Management feature.

This document will help you get started with instructions to configure the OnAlert sign to monitor and display those alert messages.

For more details on Outbound CAP format, please refer to http://docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2.html.

1.1. Applicability

This document describes the integration of the OnAlert sign firmware version 2.8 and newer only. This document will be updated with features changes implemented in future versions.

1.2. Overview: How it Works

The steps below outline the process for an outbound CAP alert to display on a Novanex LED Sign.

1. The administrator for the Alert system configures one or more CAP Outbound notification feeds to be provided at specified URLs.

2. The administrator for the OnAlert sign configures one or more Emergency Notification feeds that correspond to the specified CAP Outbound feeds.

3. The administrator for the OnAlert sign can specify the way a message is displayed by changing the alert style, presentation style, and feed element to show. Refer to Section 1.4 for more details about these options.

4. The OnAlert sign polls the specified feeds for valid CAP documents every 30 seconds.

5. A valid CAP Outbound notification will display on the OnAlert sign until one of the following criteria is met:

   - Another message is sent from the CAP system to the same feed.
   - A message is sent from the CAP system to a higher priority feed. *Note that the priority of feeds is configured as part of display configuration.*

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- The message reaches the expiration time configured as part of the CAP message.

1.3. System Requirements

The OnAlert sign and CAP feed integration requires the following:

- An operational CAP alerting system
- A network path between the OnAlert sign and the CAP alerting system server

1.4. Message Style

The formatting style of the message shown will depend on the options you select in the OnAlert sign configuration. You can select the alert style, presentation style, and feed element(s) to customize your message. Once you create and send a new valid message from the specified feed URL, the resulting text is sent to the sign face, overwriting any previous messages. *Note the exception for "Test" messages (specified via the optional "status" field in the CAP feed XML), for which the format is fixed (see "status" in Table 4: Required XML Feed Elements).*

1.4.1. Alert Style

Alert Style	Description
None	No alert tone, no amber flash
Brief	3 alert tone repetitions, 3 amber flashes
Long	10 alert tone repetitions, sufficient flashes to match the audio duration
Continuous	12 alert tone repetitions, continuous sequence of flashes as long as the message plays
Flash_LEDs	Flash all of the LEDs on the face of the sign 5 times between each iteration of a message; no alert tone or auxiliary strobe flash.

Table 1: Message Alert Styles

1.4.2. Presentation Style

Presentation Style	Description
Vertical-hold	Scroll up, medium speed, 2 sec hold per line, red color, bold font weight; headline and description separated with a colon
Vertical	Scroll up, fast speed, red color, normal font weight; headline and description separated with a colon
Vertical-slow	Scroll up, slow speed, red color, normal font weight; headline and description separated with a colon

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Presentation Style	Description
Horizontal	One of these styles, depending on the feed element(s) you select: • *For headline only or description only*: full height horizontal scroll, right to left, in red text (or yellow, for a test message) • *For both (headline and description):* The headline will show in appear mode centered on top line *(only 14 characters will be visible because this line doesn't scroll)* in red (or yellow, for a test message); the description will horizontally scroll at medium speed in yellow
Horizontal-hold	One of these styles, depending on the feed element(s) you select: • *For headline only or description only*: full height horizontal scroll, right to left, in red text (or yellow, for a test message). • *For both (headline & description):* the headline will show in appear mode centered on top line *(note: only about 14 characters will be visible because this line doesn't scroll)* in red (or yellow, for a test message), hold for as long as the description plays; the description will horizontally scroll on the bottom line at fastest speed in yellow; designed for longer messages than horizontal style

Table 2: Message Presentation Styles

1.4.3. Feed Element

You can choose to show the headline only, description only, or both. Note that if you select the horizontal presentation style, your message will display differently, depending on which feed element(s) you select.

Vertical-hold, vertical, and vertical-slow will display the same for all three feed element selections. Refer to Table 2 for details about how the horizontal presentation styles will look for the feed element choices.

1.5. Limitations

The CAP Feed and OnAlert sign have certain restrictions on message content and configuration. Consider the following limitations when configuring your OnAlert sign and CAP Feed integration:

- The OnAlert sign truncates a message after 3162 characters. This count includes the message Headline, a colon character, a space, and the Description field.
- The OnAlert sign will remove any carriage returns or line feeds from the text.
- The OnAlert sign doesn't support parentheses in the feed URL.

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- The sign will convert double-quotes to single-quotes; in order to show double-quotes in an alert message on the sign, enter two double-quotes or two single-quotes next to each other.

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1.6. Best Practices

1.6.1. Best Practice #1: Feed Priority Configuration

The OnAlert sign can monitor two feeds in the priority order shown in Table 3.

Feed	Priority level	Typical Use
Feed A	Higher priority	Feed URL for a specific building
Feed B	Lower priority	Feed URL for the entire campus or a quadrant if this sign is part of the emergency response plan

Table 3: Feed Priority Levels

Using both of the feeds in Table 3 is recommended, but not required. Only one message will be shown on the OnAlert sign at a time, the currently valid message on the highest priority feed.

1.6.2. Best Practice #2: Test vs Emergency Feed

As a best practice, you should consider configuring at least two channels at the feed provider.

- A *test feed* for a single test sign (e.g., in your dispatch office). This test feed will allow dispatchers to send shift change or training messages that can be viewed locally in the dispatch office, encouraging frequent use and familiarity with the system.
- A *campus-wide emergency notification feed* for your emergency communications to be sent to all other signs.

More details about how to configure these feeds are outlined in the following sections.

1.7. Troubleshooting

If your access to the alerting system features limits your ability to send messages, contact your alerting system administrator to request changes to the privileges.

If you follow these instructions and your sign is not properly showing messages, log in to the sign and use the *Config* command to see the current configuration. Review the configuration settings to make sure it has the proper URL(s).

Refer to the *OnAlert sign Installation and User Guide* or to the FAQs on the support site (https://support.novanexsolutions.com) for more assistance.

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2. Configure a CAP Outbound Alert

Complete the steps below to configure your emergency notification channel.

2.1. Set Up a CAP Feed

Per CAP requirements, configure your CAP alerting system to provide XML documents at the specified URL, and containing, at a minimum, the fields listed in Table 4. *Note that some of these fields are specified within the "info" element of the XML document.*

Field	Required?	Description
identifier	Yes	Only used in logging the processing of the CAP document retrieved from the Alert system server, but must be provided in the CAP feed XML.
status	No	If status is set to "Test" the message specified in the feed will be formatted as a test message (displayed text is all yellow with no siren or flash, regardless of the formatting specified in the display configuration)
msgType	Yes	Only when msgType is set to "Alert" will the message specified in the feed be displayed
sent, effective	One or the other is required	The time at which the CAP notification was sent; these fields are used to determine the time after which a message should be considered valid. If both are provided, the *effective* value is used as the authoritative time before which the message is not to be displayed.
expires	No	The time after which the CAP notification is considered invalid and should not be displayed; if not provided, the message stops playing on the sign eight hours after the sent or effective time.
headline	No	An optional part of the message; however, either the headline or description should be provided.
description	No	An optional part of the message; however, either the headline or description should be provided.

Table 4: Required XML Feed Elements

Note that the date-time format should conform to the Oasis CAP standard, which at the publishing of this document follows https://www.w3.org/TR/xmlschema-2/#dateTime.

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2.2. Example CAP Feed XML

An example CAP feed XML is presented in Figure 1; the highlighted fields are used by the OnAlert sign as described in Table 4, above, in processing the CAP feed document.

```
<alert xmlns="urn:oasis:names:tc:emergency:cap:1.2">
<identifier>14995548000026474538</identifier>
<sender>Novanex, Inc.</sender>
<sent>2018-01-24T12:00:00-05:00</sent>
<status>Actual</status>
<msgType>Alert</msgType>
<scope>Public</scope>
<info>
<language>en-US</language>
<category>Other</category>
<event>OnAlert Display Test Message</event>
<urgency>Immediate</urgency>
<severity>Extreme</severity>
<certainty>Observed</certainty>
<effective>2018-01-24T12:00:00-05:00</effective>
<onset>2018-01-24T12:02:00-05:00</onset>
<expires>2018-01-25T19:02:00-05:00</expires>
<senderName>Inova Solutions</senderName>
<headline>test headline</headline>
<description>test description</description>
</info>
</alert>
```

Figure 1: Example CAP Feed XML Document

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3. Configure the Novanex LED Sign

Once you have configured the CAP feed in your alerting system, you will need to configure the OnAlert sign to retrieve those messages. The SSH command line interface session described below is shown in Figure 2. When you are entering configuration settings in *config edit*, the existing settings are shown in parentheses before the colon of each prompt.

1. Follow the instructions for logging into the sign with an SSH client program, the admin username, and the appropriate password.
2. Type *config edit*.
3. Type 2 and press *Enter* to keep the Network settings and continue.
4. Type 2 and press *Enter* to keep the Timekeeping settings and continue.
5. Type 2 and press *Enter* to keep the Display Settings.
6. Type *1* and press Enter to edit the Emergency Communications settings.
7. Set the EC Service Enabled field to *true*.
8. Set the EC Service Provider field to *cap*.
9. Set the Feed A URL field to the file location URL shown in the CAP Channel you previously created. (http://your.server/yourCAPfeed-A in this example).
10. Set the Feed B URL field to the file location URL shown in the CAP Channel you previously created. (http://your.server/yourCAPfeed-B in this example).
11. Set the Alert style field to the desired setting: *none*, *brief*, *long*, *continuous*, or *Flash_LEDs*. (Refer to Section 1.4.1).

    If a lower priority feed was configured, set its alert style: "", *none*, *brief*, *long, continuous*, or *Flash_LEDs*.

    Use "" to set the same alert style as the main feed.
12. Set the Presentation Style to the desired setting: *vertical*, *vertical-hold*, *vertical-slow*, *horizontal,* or *horizontal-hold.* (Refer to Section 1.4.2).
13. Set the feed elements you wish to display: *headline*, *description*, or *both*. (Refer to Section 1.4.3).
14. The sign will present the Emergency Communications settings again for your review. If you are satisfied, press 2 to save your settings.
15. Wait for the command prompt to return before exiting the sign SSH session to ensure your changes were saved.

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```
Emergency Communication Settings:
  EC Service Enabled : false

1-edit Emergency Communication Settings, 2-keep and continue, 3-quit > 1

Configure Emergency Communication Settings :
  Set EC Service Enabled ( false ): true
  Set EC Service Provider ( "" ): cap
  Set CAP Feed A URL  ( "" ): http://your.server/yourCAPfeed-A
  Set CAP Feed B URL  ( "" ): http://your.server/yourCAPfeed-B
  Set Alert style( NONE ): long
  Set Alert style of low priority feed(if different)( NONE ):     brief
  Set Presentation Style ( VERTICAL-HOLD ): vertical
  Set EC feed element(s) to display in messages ( DESCRIPTION ): headline


Emergency Communication Settings:
  EC Service Enabled : true
  EC Service Provider : CAP
  CAP Feed A URL  : http://your.server/yourCAPfeed-A
  CAP Feed B URL  : http://your.server/yourCAPfeed-B
  Alert style : LONG
  Alert style of low priority feed (if different) : BRIEF
  Presentation Style : VERTICAL
  EC feed element(s) to display in messages : HEADLINE


1-edit Emergency Communication Settings, 2-keep and continue, 3-quit >2
```

Figure 2: OnAlert Sign Configuration for Monitoring a CAP Alerting System

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4. Novanex Support Site Message Management

The Novanex support site has the capability to send CAP messages to the OnAlert sign for authorized organizations. Only support site users associated with your organization can send CAP messages to your signs. Navigate to the "My Displays" page at the support site and click on the Novanex Messages link in the Message Management section.

The CAP Messages page lists recent messages and offers features for creating new messages, deleting old messages, and resending an existing message. Below the messages panel this page provides the Messaging Feed URL for your organization's messages. Enter that URL as one of the feed URLs in the OnAlert sign configuration. Then create a test message and verify that it plays on the Novanex LED Signs so configured.

5. Send a Test Message to the Sign

To verify successful configuration, send a test message from the Alerting System to the Novanex LED Sign. You can use whatever procedural steps your Alerting System requires to send a message on the URL specified for the CAP feed in the OnAlert sign configuration.

The message should appear on the OnAlert sign within 60 seconds.

6. Deactivating a Message on the Novanex LED Sign

The OnAlert sign will deactivate a CAP message when one of the following criteria is met:

- The passage of time results in the message expiring based on the timing described in the message activation criteria.
- When a new CAP message is received at an equal or higher priority feed, and meets the message activation criteria, then any previously playing message is removed, even if the new CAP message does not have any contents for the specified elements to show (i.e., headline and/or description are empty). *Note that this latter condition is incumbent on the feed operator to address.*

7. Loss of Network Connectivity at the Novanex LED Sign

The OnAlert sign polls the configured feed URLs every 30 seconds. If there is a problem connecting to the feed URLs the OnAlert sign will continue to show a currently playing message regardless of its expiration criteria.

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